Food Nutrition Estimation from a Single RGB Image

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Abstract

This report presents our final team system for nutrition estimation from a single RGB meal image. The pipeline combines YOLOv8-seg masking, MiDaS depth, a 9D geometric descriptor, direct nutrient regression (phase 4), and weight-first regression with nutrient conversion (phase 6), with MC Dropout for uncertainty. On the 326-dish test split, phase 4 achieves calorie MAE/RMSE 119.79/158.69 and improves calorie MAE by 74.46 over the matched phase-3 baseline; phase 6 obtains weight MAE/RMSE 98.9 g/152.2 g with derived calorie MAE 133.8. The phase-5 EfficientNet-B0 classifier reaches 85.86% top-1 and 97.03% top-5, and the integrated system is deployed in Gradio with editable portions and confidence-aware outputs.

1. Introduction

Automated food nutrition estimation remains difficult because visual appearance is not a direct proxy for mass or composition. Portion size can change significantly with camera distance and plate arrangement, and dishes with similar color or shape can have very different energy density.

Our final team goal is practical single-image estimation of calories, fat, protein, and carbohydrates with uncertainty reporting. The project repository is:

https://github.com/bryanchau11/csc8851_spring26_finalProject

We implemented a multi-phase system with graceful fallback when checkpoints are missing. The final integrated mode ("full") combines phases 4, 5, and 6 in one app.

Main contributions.

- A depth-aware geometric feature extractor from segmentation masks and MiDaS predictions.
- Two complementary nutrient paths: direct macro regression and weight-first conversion.
- A food-type classifier used for top-k dish labels and USDA density prior retrieval.
- End-to-end uncertainty propagation with MC Dropout and user-facing confidence display.
- A production-style Gradio interface for image upload, nutrient prediction, and ingredient editing.

2. Related Work

Prior approaches generally fall into three categories.

Image-level food recognition. Food-101 and related classification benchmarks motivate transfer learning with CNN backbones such as EfficientNet [2, 3]. These methods identify dish category well but do not directly solve portion estimation.

Nutrition regression from meal images. Nutrition5K-style benchmarks frame calorie/macronutrient prediction from meal photos [1]. Direct RGB-to-nutrients models can work, but they may conflate food identity and quantity, reducing robustness under viewpoint changes.

Segmentation and geometric cues. Modern instance segmentation (YOLOv8) and monocular depth estimators (MiDaS) provide useful structure for volume-related features from one view [4, 5]. Our method uses these signals to improve amount estimation before nutrient conversion.

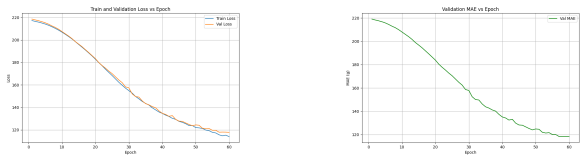
3. Dataset and Experimental Protocol

3.1. Data and Targets

Our nutrition pipeline is built around Nutrition5K-derived records containing meal-level nutrient labels:

$$\mathbf{y} = [\text{calories}, \text{fat}, \text{protein}, \text{carb}]. \tag{1}$$

Equation (1) defines the four meal-level targets. The final phase-4 notebook evaluation logs a split of 2597 train / 324 val / 326 test dishes. We report MAE and RMSE as primary metrics and include MAPE only as a secondary sig-



Loss MAE

Figure 1. Epoch-level training curves from the GPU training stage. The plots show loss and MAE across training, making the convergence trend and generalization gap visible to readers.

nal, because near-zero nutrient labels can inflate percentage errors.

3.2. Phased Development

The codebase follows six phases:

- **Phase 3**: ResNet-50 direct nutrient baseline.
- **Phase 4**: geometric-feature MLP direct regression.
- **Phase 5**: EfficientNet-B0 food classifier (Food-101).
- **Phase 6**: weight-first MLP with nutrient conversion constants.

Phases 1–2 focus on data exploration and feature engineering; phase 7 handles optional ingredient segmentation training artifacts.

Training was compute-intensive, so major jobs were executed on Kaggle GPU instances. Classifier/segmentation training and ablation reruns required multi-hour sessions before checkpoint selection, and final weights were exported for local inference in the Gradio app. Figure 1 marks where epoch-level training curves should be inserted.

4. Method

4.1. Pipeline Overview

Given RGB image I, we estimate a mask M and depth map D. A geometric encoder computes feature vector $g \in \mathbb{R}^9$. Two nutrient paths are then computed:

$$\hat{\mathbf{n}} = f_\theta(g), \tag{2}$$

$$\hat{w} = h_\phi(g), \qquad \hat{n}_j = \hat{w}\rho_j. \tag{3}$$

Equation (2) is the direct nutrient regression path, and Eq. (3) is the weight-first conversion path with per-gram constants ρ_j. Figure 2 illustrates the full system data flow, and Table 1 summarizes each module used during final inference.

4.2. Feature and Model Details

The app implementation in `app/core\_models.py` defines the phase-4 and phase-6 MLPs as:

- Input 9 features
- Hidden layers: 128 → 64 → 32
- BatchNorm + ReLU + Dropout (0.2)
- Output: 4 nutrients (phase 4) or 1 scalar weight (phase 6)

Table 1. Final integrated pipeline components and their outputs. This table is self-contained and maps each stage to the exact artifact consumed by downstream modules during inference.

Stage	Model	Output
Segmentation	YOLOv8n-seg	Food mask(s)
Depth	MiDaS$_{small}$	Relative depth map
Direct regression	NutritionMLP	$\hat{\mathbf{n}}$
Weight-first	WeightMLP	$\hat{w}$ then $\hat{\mathbf{n}}$
Dish type	EfficientNet-B0	Top-k label probs
External prior	USDA cache/API	food-specific density

The phase-5 classifier uses EfficientNet-B0 features with a dropout-MLP head for 101 Food-101 classes.

4.3. Uncertainty Modeling

At inference, dropout remains active for T stochastic forward passes (typically $T = 30$ in app inference and $T = 50$ in notebook evaluation). For prediction samples $\{\hat{y}^{(t)}\}_{t=1}^{T}$:

$$\mu = \frac{1}{T}\sum_t \hat{y}^{(t)}, \tag{4}$$

$$\sigma^2 = \frac{1}{T-1}\sum_t \left(\hat{y}^{(t)} - \mu\right)^2. \tag{5}$$

Equations (4) and (5) are the Monte Carlo mean and sample variance estimates [6]. We report confidence as approximately $\mu \pm 2\sigma$.

For weight-first macros, uncertainty is propagated linearly:

$$\sigma_{n_j} = \rho_j \sigma_w. \tag{6}$$

Equation (6) is also used in the app confidence display.

4.4. App Integration and Fallback Logic

The runtime mode resolver supports multiple checkpoint combinations:

- `full`: phase 6 + phase 4 + phase 5 loaded
- `phase6`: weight-first only
- `phase4`: direct regression only
- `phase3`: ResNet fallback

When food classification confidence is low, the app avoids overconfident labels via a minimum confidence threshold. If USDA dish-specific constants exist, they override generic constants from the local metadata cache and FoodData Central lookup [7].

4.5. Deployment Behavior and Human-in-the-Loop Editing

The deployed Gradio interface supports post-inference correction: users can edit ingredient portions (amount/unit), and the backend recomputes totals using USDA per-food densities when available, otherwise fallback dish-level constants.

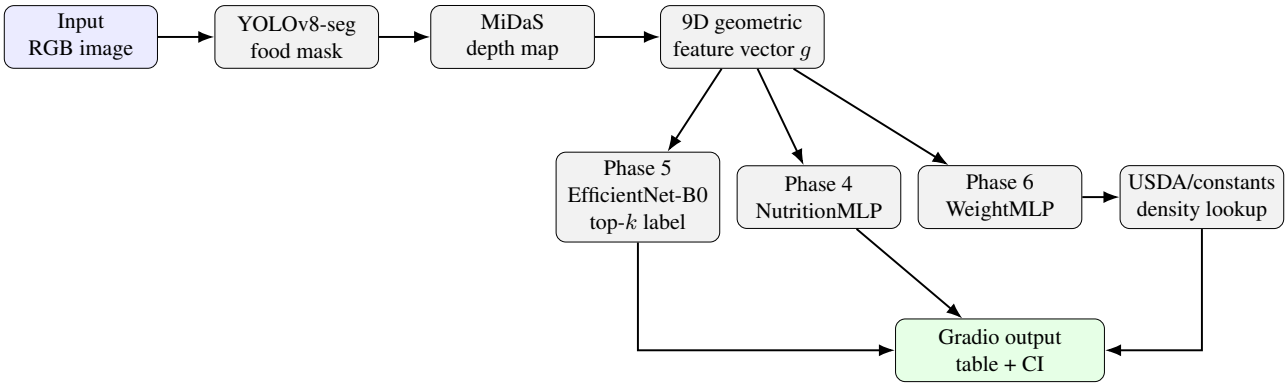


Figure 2. Final integrated inference flow in our team system. Starting from one RGB image, YOLOv8-seg generates a food mask, MiDaS provides relative depth, and a 9D geometric descriptor is sent to phase-4 (direct nutrient regression), phase-6 (weight-first prediction), and phase-5 (food classification). The Gradio layer merges these outputs with USDA/constants lookup and reports nutrients with confidence intervals.

Table 2. Phase 3 vs Phase 4 MAE on the matched 326-dish test split. Lower values indicate better accuracy; Δ MAE is computed as (Phase 4 minus Phase 3), so negative values indicate improvement from the geometric phase-4 model.

Target	P3 MAE	P4 MAE	Δ MAE
Calories	194.25	119.79	-74.46
Fat	12.79	7.67	-5.13
Protein	19.37	11.01	-8.36
Carb	16.01	11.14	-4.87

To improve safety and interpretability, the app exposes intermediate quantities (dish label, grams, confidence, per-item calories) and suppresses strong dish assertions when classifier confidence is below threshold.

5. Experiments and Results

All stage comparisons use the same fixed split (2597 train / 324 val / 326 test) and matched-split MAE/RMSE reporting to avoid protocol drift.

5.1. Phase 3 to Phase 4 (Matched Split Comparison)

The phase-4 notebook reports a direct phase-3 vs phase-4 MAE comparison on the same 326-dish split (Table 2).

5.2. Phase 4 Direct Nutrition Metrics

Table 3 reports the complete phase-4 test-set MAE/RMSE summary obtained with 50 MC-dropout samples per dish.

5.3. Phase 6 Weight-First Metrics

Phase-6 test output reports:

- Weight MAE/RMSE: 98.9 g / 152.2 g
- Mean actual test weight: 192.6 g
- Relative error: 51.4%

Table 3. Phase-4 test metrics with MC Dropout (50 samples). Units are kcal for calories and grams for fat/protein/carbohydrates.

Target	MAE	RMSE
Calories	119.79	158.69
Fat	7.67	10.66
Protein	11.01	15.75
Carb	11.14	14.35

Table 4. Phase-6 derived nutrient MAE from the weight-first path. Nutrient values are computed as predicted weight multiplied by learned per-gram constants; calorie MAE is in kcal and macronutrient MAE values are in grams.

Metric	Value
Derived calorie MAE	133.8
Derived fat MAE	8.1
Derived protein MAE	11.9
Derived carb MAE	10.9

and derived macro MAE values shown in Table 4. The stored checkpoint metadata records a best validation weight MAE of 106.59 g.

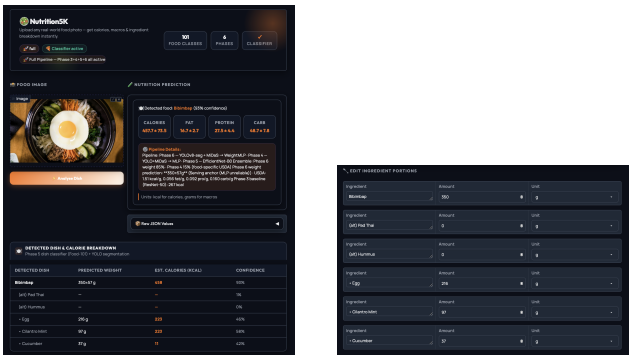
5.4. Phase 5 Classification

From `models/best_food_classifier.pt`:

- Validation top-1 = 0.8586 (85.86%)
- Validation top-5 = 0.9703 (97.03%)
- Best checkpoint epoch = 23

Figure 3 shows qualitative screenshots from the deployed app.

The weight prediction detail panel is summarized in text rather than a screenshot. It explains how the 9D geometric features are formed from YOLO mask area and MiDaS depth



Result panel Ingredient editing panel

Figure 3. Qualitative screenshots from the deployed Gradio app, showing the result panel and ingredient edit interface.

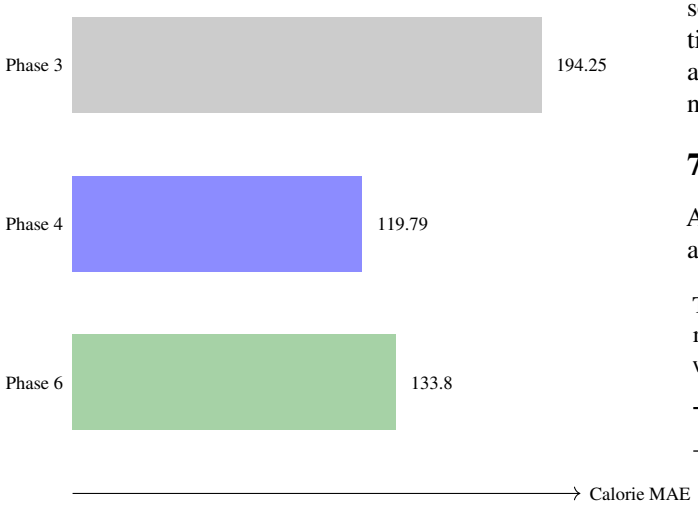


Figure 4. Calorie MAE comparison across stage outputs in the final system. Phase 4 yields the lowest error, while Phase 6 remains competitive and offers gram-level interpretability via the weight-first path.

statistics, how MC-dropout is used to estimate uncertainty, and how the final gram estimate is converted into calories and macros with dish-specific constants when available.

5.5. Visual Comparison Across Stages

Figure 4 summarizes calorie MAE across the three available stage outputs. The largest gain is phase 4 over the phase-3 baseline, while phase 6 remains competitive and adds interpretable gram-level outputs.

5.6. Analysis

The most reliable quantitative gain is phase-4 improvement over phase 3 on the matched split. Phase 6 remains useful for interpretability because users can inspect predicted grams and explicit nutrient conversion.

Largest errors appear in mixed dishes, strong occlusion/viewpoint changes, and foods with similar geometry but different density. In deployment, uncertainty is reported as $\mu \pm \sigma$ and $\mu \pm 2\sigma$ to support safer interpretation (Figure 5).

Figure 5. Uncertainty interval concept used in app reporting.

6. Conclusions

This team project produced an end-to-end nutrition estimation system that is practical and extensible. The integrated pipeline combines segmentation, depth, lightweight regressors, uncertainty reporting, and food-prior retrieval in a real-time Gradio demo. Future work will expand dish coverage and improve calibration between classifier confidence and nutrient priors.

7. Peer Evaluation

All three teammates contributed to the final project pipeline and report; Table 5 summarizes responsibility allocation.

Table 5. Team contribution summary for peer evaluation. Shares reflect primary ownership of implementation, experiments, and writing tasks.

Member	Primary responsibilities	Score (/20)
Bryan Chau	Integration, app, final report	20
Loc Giang	Data/model experimentation	19
Katelyn Truong	Classification/evaluation support	19

References

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[7] USDA, "FoodData Central API Guide," https://fdc.nal.usda.gov/api-guide, accessed 2026. 2